Outline of the topic and research project

The geometry of subvarieties on abelian varieties

Complex abelian varieties provide a setting in which algebraic geometry interacts closely with topology and representation theory. The aim of this doctoral project is to understand the geometry of their subvarieties through Gauss maps and conormal cycles. Particular attention will be paid to the irreducibility, dimension, and homology classes of subvarieties obtained by adding points in fibres of the conormal Gauss map. These constructions give a geometric approach to cycles arising from the convolution of perverse sheaves and the associated Tannaka groups.

For a subvariety of an abelian variety, the tangent Gauss map records how its tangent spaces vary after translation to the origin. The conormal Gauss map records the invariant covectors annihilating these tangent spaces. When the latter map is generically finite, its monodromy describes the permutations of the points in a general fibre. Taking weighted sums of these points produces new subvarieties, whose geometry depends on the monodromy action. The project studies both this action and the geometric invariants of the resulting subvarieties.

The work completed so far establishes a geometric framework for these constructions and provides monodromy calculations for curves in Jacobian and Prym varieties. It also gives explicit formulas for the Chern-Mather classes of the constructed subvarieties when the conormal monodromy is the full symmetric group, or, in the symmetric case, the full signed symmetric group. The formulas combine Pontryagin products with combinatorial coefficients obtained from partitions. They allow dimensions and fundamental homology classes to be recovered and yield concrete descriptions in the Jacobian case.

Recent counterexamples require a refinement of the broader conjectural picture. A curve in a simple abelian threefold can have a tangent Gauss map of degree three without any nontrivial ambient automorphism preserving the curve and acting by a scalar on tangent spaces. A separate construction involving Schoen surfaces shows that the two half-spin cycles associated with type D monodromy can have different homology classes. These examples reveal geometric information not captured by the previously proposed symmetry principles and provide starting points for further investigation.

The next stage has three connected objectives. First, I will seek sufficient geometric conditions for generic injectivity of the tangent Gauss map and for large conormal monodromy, using linear systems on curves, Jacobian quotients, and Prym constructions. Second, I will investigate how the existing characteristic-class formulas can be extended to type D monodromy while retaining the distinction between the two half-spin orbits. Third, I will use explicit examples and degenerations to study which features determine the dimensions and homology classes of the resulting supports. The question of equality of the half-spin support dimensions will be treated separately from the disproved equality of their homology classes.

The methods combine the geometry of curves and abelian varieties, covering-space monodromy, intersection theory, and the combinatorics of Weyl group actions. Explicit symbolic calculations will support the analysis of examples. The intended outcome is a collection of precise criteria and computable invariants that explains both the established positive results and the newly identified exceptional behaviour.